

Requirements & Workload Specification Report

Cloud-Native Distributed Music Streaming Platform (Spotify Workload)

Course: Cloud Computing (Semester 2026-III)

Universidad Distrital Francisco José de Caldas

Team 6

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Abstract

This report presents the formal requirements engineering and quantitative workload modeling for a globally distributed, cloud-native music streaming and personalization platform based on the operational profile of Spotify. The system is designed to sustain over 600 million registered and active users, a catalog exceeding 100 million audio tracks, and more than 25 million average (40 million peak) concurrent audio streams. Critical operational constraints include maintaining 99.9% playback availability, delivering sub-500 ms personalized recommendations at the 95th percentile, absorbing up to 64 million events per second in telemetry ingestion, and resolving cross-device playlist state conflicts deterministically. In strict compliance with the specifications for Delivery No. 1.

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1. Executive Summary

Modern digital audio streaming platforms operate at an unprecedented scale, combining massive, read-heavy immutable content delivery with high-throughput state updates and low-latency predictive personalization. The subject of this project, **Team 6 — Spotify (Music Streaming)**, addresses the operational demands of delivering high-fidelity audio, dynamic playlist synchronization, and contextual music discovery across more than 180 countries.

The primary engineering challenges governing this system encompass:

- **High-Concurrency Media Ingress and Egress:** Managing continuous audio stream delivery for over 25 million average and 40 million peak concurrent listeners, representing an aggregate media egress bandwidth exceeding 6.4 Tbps while sustaining 99.9% playback availability (< 43.8 minutes of monthly downtime).
- **Low-Latency Personalization at Scale:** Scoring and assembling personalized recommendation feeds within a strict 500 ms threshold at the 95th percentile (p_{95}).
- **Cross-Device and Offline State Consistency:** Handling concurrent playlist mutations from multiple client sessions and reconciling offline synchronization without data loss.
- **Continuous Telemetry Ingestion:** Reliably recording and aggregating telemetry pipelines processing up to 64 million events per second during peak hours to satisfy legal royalty auditability and near-real-time user taste recomputation.

This specification formalizes the system's operational parameters, actor behaviors, functional requirements, and measurable service quality targets without committing to specific cloud service providers or software vendor technologies.

2. System Description

2.1 Business Problem

The transition from local media ownership to cloud-hosted on-demand streaming fundamentally changed how users consume audio and how creators monetize content. The platform addresses three primary business pain points:

1. **Universal Content Accessibility:** Eliminating consumer storage limitations by providing immediate access to a centralized catalog of over 100 million recorded tracks.
2. **Discovery Fatigue Mitigation:** Transforming an intractable volume of music into hyper-personalized, context-aware listening paths that maximize listener engagement and catalog breadth.
3. **Creator Rights and Royalty Accounting:** Guaranteeing non-repudiable and accurate logging of track consumption to ensure proportional royalty compensation across rights holders and publishing organizations.

From a distributed systems standpoint, the system is operationally challenging due to extreme read/write divergence. Media requests and catalog indexing require globally distributed, low-latency read caching, whereas playback beacons, telemetry, and collaborative playlist updates constitute continuous, high-volume write operations that demand deterministic conflict resolution and guaranteed eventual consistency.

2.2 Actor Catalog

The platform interfaces with both human users and external automated systems, as detailed in Table ??.

Actor Identifier	Type	Role & Description	Interaction Frequency	System Expectations
ACT-01: Free Listener	Human User	Accesses ad-supported streaming with shuffle restrictions on mobile clients.	Continuous during active daily sessions (avg. 2.4 hrs/day).	Audio initiation < 200 ms, responsive catalog search.
ACT-02: Premium Subscriber	Human User	Pays a recurring fee for on-demand playback, offline downloads, and multi-device control.	Continuous during active daily sessions (avg. 2.4 hrs/day).	99.9% availability, instant state sync, lossless offline playback.
ACT-03: Content Provider	Human / Enterprise	Record labels and independent artists uploading media, track metadata, and reviewing consumption metrics.	Periodic (daily/weekly media uploads and analytics reviews).	Ingestion data integrity, timely consumption reports, catalog protection.
ACT-04: Edge Relay (CDN)	External System	Geographically distributed nodes caching and streaming encrypted audio chunks directly to clients.	Extremely high / Continuous (millions of media segment pulls per second).	Sub-millisecond manifest validation, high edge cache hit ratios.
ACT-05: Royalty Gateway	External System	Auditing and collection societies receiving certified playback logs for license reconciliation.	Asynchronous batch processing (daily/monthly log ingestion).	Tamper-evident, deduplicated playback records (30 seconds).
ACT-06: Identity Gateway	External System	Third-party federated authentication and billing platforms handling account entitlements.	High at session initiation and periodic token refresh (hourly).	High availability, standardized token validation, secure callbacks.

Table 1: System Actor Catalog

2.3 Main Use Cases

2.3.1 UC-01: Stream Audio Track On-Demand

- **Actors:** Free Listener (ACT-01), Premium Subscriber (ACT-02), Edge Relay (ACT-04).
- **Preconditions:** The user is authenticated; the requested track is indexed in the catalog and licensed in the user's registered territory.
- **Main Flow:**
 1. The user selects an audio track to play.
 2. The client issues an authorized streaming request to the core application service.
 3. The system validates account entitlements, active playback sessions, and geographic licensing rules.
 4. The system determines the optimal bitrate and returns a signed, time-limited playback manifest.
 5. The client requests media segments directly from the nearest Edge Relay.
 6. Playback begins within 200 ms, and the client starts sending periodic playback telemetry beacons.
- **Alternative Flows:**
 - *Territorial Restriction:* If licensing prohibits playback in the detected geographic location, the system returns an error status and prompts the client to advance to the next track.
 - *Skip Limit Exceeded (Free Tier):* If a Free Listener exceeds the permitted skips per hour, the explicit selection is denied, and the client resumes algorithmic radio playback.
 - *Edge Cache Miss:* If the requested segment is not cached at the edge, the Edge Relay fetches the master segment from the core storage tier before streaming it to the user.
- **Postconditions:** Playback is successfully initiated; an active playback session is registered; telemetry is queued for royalty calculation.

2.3.2 UC-02: Fetch Personalized Recommendation Feed

- **Actors:** Free Listener (ACT-01), Premium Subscriber (ACT-02).
- **Preconditions:** The user is authenticated in the client application.
- **Main Flow:**
 1. The user opens or refreshes the application Home screen.
 2. The client requests the user's personalized recommendation feed.
 3. The system resolves active A/B testing experiment cohorts assigned to the user.
 4. The system retrieves candidate recommendations generated via collaborative filtering and taste profile models.
 5. The system scores, ranks, and deduplicates the candidate cards.
 6. The assembled feed is returned to the client within 500 ms (p_{95}).
- **Alternative Flows:**
 - *Cold-Start (New User):* If no historical listening profile exists, the system serves a curated onboarding feed based on declared genre preferences and regional popular trends.
 - *Recommendation Model Timeout:* If dynamic ranking exceeds 350 ms, the system falls back to a cached baseline feed to guarantee response deadlines.
- **Postconditions:** The client renders the feed; impression events are published to the telemetry queue.

2.3.3 UC-03: Collaborative Playlist State Synchronization

- **Actors:** Premium Subscriber (ACT-02), Free Listener (ACT-01).

- **Preconditions:** A playlist is configured with collaborative editing permissions; participating users have loaded the playlist.
- **Main Flow:**
 1. A user appends, reorders, or deletes a track within the shared playlist.
 2. The local client applies the change optimistically and transmits the mutation with logical sequence metadata to the system.
 3. The system verifies write authorization on the target playlist.
 4. The system applies deterministic conflict resolution algorithms to merge concurrent mutations.
 5. The updated state version is persistently committed.
 6. A delta update notification is broadcast to all active connected collaborator sessions.
- **Alternative Flows:**
 - *Offline Edits:* Mutations made while offline are queued locally and synchronized upon network reconnection; conflicts are resolved deterministically based on logical timestamp vectors.
- **Postconditions:** Persistent playlist state is updated; all connected client views converge to an identical ordering.

2.3.4 UC-04: Synchronize Encrypted Audio for Offline Mode

- **Actors:** Premium Subscriber (ACT-02).
- **Preconditions:** The user maintains an active Premium subscription; the device is registered within the authorized offline device quota (up to 5 devices).
- **Main Flow:**
 1. The user flags a playlist or album for offline consumption.
 2. The client requests offline authorization and cryptographic media licenses.
 3. The system validates account standing and available device slots.
 4. The system issues signed download manifests containing 30-day lease authorizations.
 5. The client downloads encrypted audio chunks and verifies local file integrity.
 6. The client acknowledges successful synchronization to the server.
- **Alternative Flows:**
 - *Device Quota Exceeded:* If the user attempts to authorize more than 5 devices, the system prompts the user to de-register an existing device before proceeding.
 - *Subscription Inactive:* If the account transitions to a free tier, all local offline licenses are revoked immediately.
- **Postconditions:** Media chunks are stored in an encrypted state on the device, playable without active network connectivity for up to 30 days.

2.4 System Boundaries

The system boundary defines the functional scope of the core distributed software platform, separating core operational responsibilities from external infrastructure, third-party services, and client hardware.

- **In-Scope:** Account entitlement validation, catalog search indexing, collaborative playlist conflict resolution, recommendation inference and scoring, playback manifest generation, dynamic A/B experimentation routing, and high-throughput telemetry aggregation.
- **Out-of-Scope:** Physical global fiber networks and edge point-of-presence hardware, direct banking transaction settlement, original audio mixing and digital-audio-workstation software, client-side digital-to-analog audio conversion, and third-party advertising exchange bidding networks.

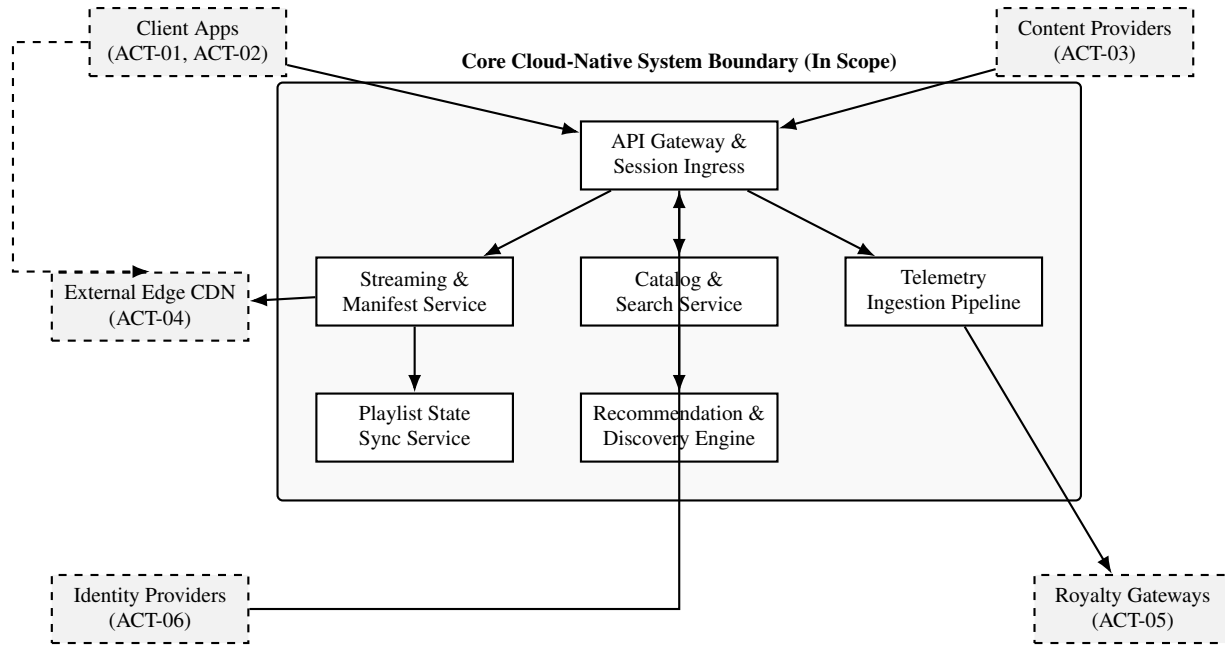


Figure 1: System Boundary Diagram

3. Functional Requirements

All functional requirements strictly follow the standardized format: *"The system shall [action] [object] [condition/constraint]"*. They are uniquely identified and categorized by service domain.

3.1 Domain 1: Streaming & Media Delivery

- **FR-01:** The system shall authenticate media playback requests and generate cryptographically signed media manifests specifying valid edge segment locations within 100 ms of request validation (Traceable to UC-01).
- **FR-02:** The system shall regulate audio streaming bitrates dynamically based on client-reported network bandwidth and buffer health metrics (Traceable to UC-01).

3.2 Domain 2: Catalog & Search

- **FR-03:** The system shall execute typo-tolerant text searches across the global catalog of over 100 million tracks, albums, and artists, returning matching results within 150 ms at the 95th percentile (Traceable to UC-01, UC-02).
- **FR-04:** The system shall enforce territorial licensing restrictions by filtering catalog track availability dynamically according to the user's verified geographic location (Traceable to UC-01).

3.3 Domain 3: Personalization & Machine Learning

- **FR-05:** The system shall generate personalized recommendation feeds and dynamic discovery carousels for authenticated users within 500 ms at the 95th percentile (Traceable to UC-02).
- **FR-06:** The system shall ingest individual user interaction telemetry (skips, track completions, repeats) and recompute user affinity representations within 10 minutes of event emission (Traceable to UC-02).

3.4 Domain 4: Playlist Management & Synchronization

- **FR-07:** The system shall synchronize concurrent modifications to collaborative playlists across all connected devices using deterministic conflict resolution algorithms to eliminate state divergence (Traceable to UC-03).
- **FR-08:** The system shall issue encrypted audio leases with a maximum 30-day offline validity period exclusively to verified Premium subscribers on registered devices (Traceable to UC-04).

3.5 Domain 5: Social Features & Real-Time Presence

- **FR-09:** The system shall broadcast playback transport state updates (play, pause, seek, track transition) to all active participants in a collaborative group session within 100 ms of the host event (Traceable to UC-03).

3.6 Domain 6: Experimentation & Telemetry

- **FR-10:** The system shall evaluate active A/B feature flags and assign users to experiment cohorts with an evaluation latency overhead of less than 5 ms per request (Traceable to UC-02).
- **FR-11:** The system shall record and persist tamper-evident, deduplicated playback audit logs for every continuous track stream reaching or exceeding 30 seconds to support royalty reconciliation (Traceable to UC-01).

4. Non-Functional Requirements

The non-functional requirements define the quantitative, verifiable constraints and performance standards of the distributed system.

- **NFR-01 (Expected Users):** The system shall support a minimum of 602 million registered monthly active users (MAU), with an active base of at least 250 million daily active users (DAU), maintaining an average of 25 million and a peak of over 40 million concurrent active playback streams.
- **NFR-02 (Average Requests/Hour):** The system shall process an average baseline volume of 1.5 billion API and metadata requests per hour ($\approx 416,667$ requests per second) during normal operational hours, excluding direct CDN edge media segment streaming.
- **NFR-03 (Peak Requests/Hour):** The system shall absorb a peak traffic volume of up to 3.75 billion API and metadata requests per hour ($\approx 1,041,667$ requests per second) during synchronized global peak release windows without request shedding or elevated error rates.
- **NFR-04 (Peak Duration):** The system shall sustain peak load operating conditions continuously for a minimum window of 4 hours without performance degradation.
- **NFR-05 (Response-Time Targets):** The system shall satisfy the following latency constraints across all operational endpoints:
 - 95th Percentile (p_{95}): < 200 ms for standard catalog retrieval, search queries, and playlist mutation requests.
 - 95th Percentile (p_{95}): < 500 ms for personalized recommendation feed scoring and assembly.
 - 99th Percentile (p_{99}): < 1.0 s for complex multi-term searches across global catalog indexes.

- **NFR-06 (Availability):** The system shall maintain an overall playback availability of 99.9% (the three-nines standard), allowing no more than 43 minutes and 49 seconds of unscheduled system downtime per calendar month.
- **NFR-07 (Data Volume at Launch):** The system shall manage an initial operational data footprint of at least 18 Petabytes (PB), comprising 100 million unique master tracks stored across multiple encoded bitrates (12.6 PB triply replicated), plus 5.4 PB in cover artwork, canvas video loops, metadata indexes, and vector embeddings.
- **NFR-08 (Data Growth):** The system shall accommodate an annual data storage growth rate of at least 5.8 PB per year, driven by the daily ingest of approximately 100,000 new releases (4.6 PB/year replicated) and 1.2 PB/year of telemetry events and cold analytical logs.
- **NFR-09 (Geographic Scope):** The system shall serve traffic across more than 180 countries and territories worldwide, enforcing an edge network round-trip time (RTT) to the client of less than 50 ms in primary metropolitan target zones.
- **NFR-10 (Read/Write Concurrency Balance):** The system shall support an asymmetric workload profile characterized by a 90% Read to 10% Write ratio across client-facing transactional core services, while scaling an independent asynchronous telemetry ingestion pipeline capable of absorbing up to 64 million event messages per second during peak events.

5. Workload Model

5.1 Structured Workload Model Table

Table 2 provides the operational parameters characterizing the target system load under normal and peak conditions.

5.2 Justification of Workload Parameters

All values in the workload model are derived from empirical disclosures, official engineering publications, and distributed systems baselines:

1. **User Base, Concurrency, and Request Throughput:** According to Spotify's SEC Form 20-F filings (2024), the platform reports over 602 million Monthly Active Users (MAUs) and 236 million paying subscribers. In media streaming, the Daily Active User to Monthly Active User ratio (DAU/MAU) averages approximately 42%, establishing an active user base of 250 million daily active listeners. According to the International Federation of the Phonographic Industry (IFPI) Global Music Report, streaming consumers listen to an average of 2.4 hours of audio per day. Across a 24-hour cycle, this translates to:

$$\frac{250,000,000 \text{ users} \times 2.4 \text{ hours/day}}{24 \text{ hours/day}} = 25,000,000 \text{ average concurrent streams} \quad (1)$$

Accounting for regional diurnal overlapping and synchronized high-profile releases, peak concurrency scales by a factor of $1.6\times$, reaching over 40 million concurrent audio streams.

Active clients generate approximately 30 metadata requests per hour (polling, playback state reporting, token validation, search queries, and browse carousels), while non-streaming active users contribute additional navigation load. This results in an average API load of 1.5 billion requests per hour ($\approx 416,667$ RPS), surging by $2.5\times$ during peak global releases to 3.75 billion requests per hour

Table 2: Quantitative Workload Model Specification

Workload Parameter	Specification Value	Verification & Calculation Basis
Average Requests/Hour (API)	1,500,000,000 req/hour ($\approx$ 416,667 RPS)	Baseline metadata, search, and heartbeat traffic across 250M DAU averaging 30 API operations per active hour.
Peak Requests/Hour (API)	3,750,000,000 req/hour ($\approx$ 1,041,667 RPS)	2.5 $\times$ surge factor driven by global album drop synchronization and concentrated evening diurnal leisure hours.
Concurrent Playback Streams	25,000,000 (Average) 40,000,000+ (Peak)	Derived from 250M DAU consuming 2.4 hours of audio daily (IFPI), peaking during synchronized multi-region windows.
Audio Egress Bandwidth	4.0 Tbps (Average) 6.4 Tbps (Peak)	Calculated at a weighted average streaming bitrate of 160 kbps (20 KB/s) across concurrent listeners.
Telemetry / Event Ingestion	Up to 64,000,000 events/sec	Empirical peak event throughput documented in Spotify Engineering's log and telemetry processing architecture.
Peak Duration	4 hours	Critical peak consumption window (Friday midnight release windows and evening commute/leisure hours).
Read / Write Ratio	90% Read / 10% Write	Asymmetric ratio for core application services (telemetry is decoupled into a dedicated append-only pipeline).
Availability Target (SLA)	99.9% (< 43.8 min/month downtime)	Industry standard Service Level Objective for commercial digital media streaming.
Response Time ($p_{.95}$)	< 200 ms (API) / < 500 ms (Recs)	Strict latency thresholds required to prevent perceived client interface lag and audio buffer starvation.
Data Volume (Launch)	18 Petabytes (PB)	100M tracks $\times$ multi-codec encodings $\times$ 3 $\times$ replication (12.6 PB) + canvas videos, artwork, vectors (5.4 PB).
Annual Data Growth	5.8 PB / year	Daily ingest of 100,000 tracks (4.6 PB/year replicated) plus 1.2 PB/year of aggregated analytical logs.
Geographic Distribution	Europe: 28%, North America: 28%, Latin America: 21%, Rest of World: 23%	Official geographic user distribution reported in Spotify SEC Form 20-F filings.

($\approx 1,041,667$ RPS). Furthermore, at a weighted average audio streaming bitrate of 160 kbps (20 KB/s across standard 96k, 160k, and 320k streams), audio egress bandwidth requires 4.0 Tbps under average conditions and 6.4 Tbps during peak hours, necessitating edge CDN offloading.

2. **Telemetry and Event Delivery Pipeline:** Beyond transactional client API requests, playback event tracking (30-second billable play milestones, interaction skips, volume adjustments, impressions) generates massive streaming event volumes. Spotify R&D’s published architectural benchmarks (*Event Delivery at Spotify*) confirm peak telemetry ingestion rates exceeding 64 million events per second globally, which are queued and decoupled from transactional user databases.
3. **Peak Window Duration:** Media streaming consumption concentrates heavily during regional leisure windows (18:00 to 22:00 local time) and persists across global time-zone boundaries during midnight releases. A 4-hour sustained peak endurance specification ensures that autoscaling thresholds, thread pools, and distributed message queues avoid memory leak exhaustion or systemic degradation under prolonged maximum pressure.
4. **Read/Write Ratio:** In consumer-facing audio streaming, user interaction is overwhelmingly read-dominated. A user inspects dozens of playlists, browse recommendations, and metadata cards before executing an explicit write operation (e.g., modifying a playlist, following an artist, or adjusting preferences). This establishes a 90% Read to 10% Write ratio across transactional core services. Write-heavy operational telemetry is isolated within the dedicated append-only ingestion pipeline.
5. **Availability and Response Latency:** Auditory perception research demonstrates that playback initiation delays exceeding 200 ms induce perceived lag and interface stuttering. Consequently, signed manifest issuance and metadata endpoints are bounded to 200 ms (p_{95}). Personalized recommendation scoring involves multi-candidate collaborative filtering across massive vector indices and is budgeted up to 500 ms at p_{95} . An availability standard of 99.9% represents an optimal Service Level Objective for consumer entertainment, permitting no more than 43 minutes and 49 seconds of unplanned monthly downtime without the exponential infrastructure cost of five-nines safety-critical systems.

6. Storage Footprint and Annual Growth:

- *Encoded Audio Media:* The catalog encompasses 100 million unique master tracks with an average duration of 3.5 minutes. Encoding across multiple operational profiles (Ogg Vorbis 96k, 160k, 320k; AAC 96k, 160k, 320k; and uncompressed 16-bit/44.1 kHz FLAC masters) requires approximately 42 MB per track in aggregate:

$$100,000,000 \times 42 \text{ MB} = 4.2 \text{ Petabytes (PB) of raw audio media} \quad (2)$$

- *Replication and Redundancy:* Enforcing triple geographic replication ($3\times$) across primary data regions scales this to 12.6 PB.
- *Artwork, Canvas Video Loops, and Embeddings:* High-resolution album artwork (multiple mipmap resolutions), 8-second looping canvas videos for top tracks ($\approx 25\text{M}$ videos at 2.5 MB each ≈ 62.5 TB), user vector embeddings, search indices, and relational database state contribute an additional 5.4 PB, establishing an initial operational baseline of **18 PB**.
- *Annual Data Expansion:* Industry analytics from Luminate and IFPI document approximately 100,000 new tracks uploaded globally each day. Encoded across all formats with triple replication, new media adds 4.6 PB/year ($100,000 \times 42 \text{ MB} \times 3 \times 365$). Retained analytical telemetry, compliance logs, and model checkpoints contribute an additional 1.2 PB/year, resulting in an annual storage growth rate of **5.8 PB/year**.

6. References

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